

Project Planning & Management

1. Project Proposal

- **Project Title:** Predictive Maintenance System Using Machine Learning
- **Overview:** A ML-based system that predicts whether a machine requires maintenance within the next 7 days based on sensor data such as temperature, vibration, and pressure readings...etc
- **Objectives:**
 - Build a reliable binary classification model for maintenance prediction
 - Provide early warnings to reduce unexpected machine downtime
 - Enable data-driven maintenance scheduling
- **Scope:** The system covers data collection from sensors and prediction output. Not covering the physical maintenance execution or hardware integration.

2. Project Plan

Phase	Task	Deadline
Planning & Management	Proposal, plan, roles, risks	2/20/2026
Literature Review	Research & evaluation	4/20/2026
Requirements Gathering	Stakeholders, user stories, requirements	4/20/2026
System Analysis & Design	Architecture, DB design, diagrams	5/1/2026
Implementation	Source code & execution	7/10/2026
Final Presentation	Reports, testing, demo	7/17/2026

Milestones:

- Train and validate ML model
- Deployed prediction API
- Completed documentation and final presentation

3. Task Assignment & Roles

Role	Responsibilities
ML Engineer	Model development, training, evaluation, and retraining pipeline
Backend Developer	API development, model serving, and data logging
Frontend Developer	User interface for inputting sensor data and displaying predictions
Data Engineer	Data collection, preprocessing, and pipeline management
Project Manager	Coordination, documentation, deadline tracking, and GitHub management

4. Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Imbalanced dataset (few cases)	High	High	Apply oversampling (SMOTE)
Model overfitting	Medium	High	Use cross-validation and regularization
Sensor data missing or noisy	Medium	High	Implement data validation and imputation
Team member unavailability	Medium	Medium	Distribute tasks with backup assignments
Deployment failures	Low	High	Test in staging environment before production

5. KPIs (Key Performance Indicators)

KPI	Target
Model Accuracy	$\geq 90\%$
Precision & Recall (Maintenance Class)	$\geq 80\%$
Prediction Response Time	< 2 seconds
System Uptime	$\geq 99\%$
False Negative Rate	As low as possible
GitHub Commit Activity	Regular commits from all team members which later approved by leader